

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1505

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I Saranya.R.A(192572052) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Saranya.R.A

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Q1.Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

ANSWER :

Introduction :

A hypervisor is virtualization software that allows multiple virtual machines (VMs) to run on a single physical server. There are two main types: Type-1 (Bare-metal) and Type-2 (Hosted). Choosing the correct hypervisor is important for mission-critical applications because these applications require high performance, strong security, and reliable management.

Analysis :

Performance :

Type-1 Hypervisor

- Runs directly on the physical hardware.
- Has very little overhead.
- Provides faster CPU, memory, and storage performance.
- Suitable for enterprise cloud datacenters.

Type-2 Hypervisor

- Runs on top of an operating system.
- Additional OS layer increases overhead.
- Performance is slower compared to Type-1.
- Suitable mainly for testing and development.

Analysis: Since mission-critical applications require high-speed processing and low latency, Type-1 performs better.

Security :

Type-1

- Smaller attack surface.
- Fewer software layers reduce vulnerabilities.
- Better VM isolation.
- More secure for sensitive business applications.

Type-2

- Depends on the host operating system.
- If the host OS is compromised, all VMs are at risk.
- Larger attack surface.

Analysis: Organizations handling financial or healthcare data require stronger security, making Type-1 the better option.

Manageability :

Type-1

- Supports centralized management.
- Easy VM migration and resource allocation.
- High availability and disaster recovery features.

Type-2

- Easy to install.
- Best for individual users.
- Limited enterprise management features.

Recommendation :

A cloud datacenter hosting mission-critical applications should use a **Type-1 Hypervisor** because it provides:

- Better performance
- Stronger security
- Better scalability
- Easier centralized management

Examples include VMware ESXi, Microsoft Hyper-V, and Xen.

Conclusion :

Type-1 hypervisors are the best choice for enterprise cloud environments because they provide high performance, strong security, and efficient management. Type-2 hypervisors are more suitable for personal use, software testing, and development.

Q2.A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

ANSWER :

Introduction :

A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Analysis :

How Virtualization Provides Isolation

- Every department gets its own virtual machine.
- Each VM has separate CPU, memory, storage, and network resources.
- One VM cannot directly access another VM's data.
- Hypervisor controls communication and resource allocation.
- Failure of one VM does not affect other VMs.

Example:

Finance VM → Financial records

Healthcare VM → Patient information

Student Portal VM → Student database

All three run on the same physical server but remain isolated.

Attack Surface 1: Hypervisor Attack

If an attacker compromises the hypervisor,

- Every VM may become accessible.
- Sensitive information from all departments can be exposed.

Protection:

- Regular hypervisor updates
- Strong administrator authentication
- Security patches

Attack Surface 2: VM Escape

VM Escape is an attack where malicious software inside one VM escapes into the hypervisor or another VM.

Consequences:

- Unauthorized access
- Data theft
- Service disruption

Protection:

- Strong VM isolation
- Updated virtualization software
- Access control policies

Recommendation :

The cloud provider should:

- Use a secure Type-1 hypervisor.
- Apply strict network segmentation.
- Enable VM isolation.
- Monitor VM activities continuously.
- Use firewalls and intrusion detection systems.

Conclusion :

Virtualization architecture allows multiple departments to safely share hardware while maintaining data isolation. Securing the hypervisor and preventing VM escape attacks are essential for protecting sensitive workloads.

Q3.Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

ANSWER :

Introduction :

The given VM host statistics indicate that the server is under heavy load. High resource utilization affects VM performance and user experience.

Analysis :

CPU Utilization = 92%

Analysis:

- CPU is almost fully utilized.
- Applications may respond slowly.
- VMs compete for processor time.

Likely
CPU overload.

Bottleneck:

Corrective Action:

- Distribute VMs across hosts.
- Increase CPU resources.
- Enable load balancing.

Memory Utilization = 88%

Analysis:

- Memory is nearly exhausted.
- System may begin swapping.
- VM performance decreases.

Likely
Memory shortage.

Bottleneck:

Corrective Action:

- Increase RAM.
- Optimize memory allocation.
- Remove unused VMs.

Storage I/O Latency = 35 ms

Analysis:

- Disk response is slow.
- Applications wait longer for data.
- Database performance decreases.

Likely
Storage subsystem.

Bottleneck:

Corrective Action:

- Replace HDD with SSD/NVMe.
- Optimize storage configuration.
- Use faster storage arrays.

VM Boot Time = 170 seconds

Analysis:

- VM startup is much slower than expected.
- Indicates CPU, memory, or storage bottlenecks.

Corrective Action:

- Reduce startup load.
- Optimize VM configuration.
- Improve storage performance.

Recommendation :

The cloud administrator should:

- Add CPU cores.
- Upgrade RAM.
- Use SSD/NVMe storage.
- Balance workloads across hosts.
- Continuously monitor system performance.

Conclusion :

The server is experiencing CPU, memory, and storage bottlenecks. Hardware upgrades, load balancing, and resource optimization will significantly improve VM performance and reduce boot time.

Q4. Analyze how Software Defined Datacenter (SDDC) architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

ANSWER :

Introduction :

A Software Defined Datacenter (SDDC) virtualizes compute, storage, and networking resources, allowing them to be managed through software instead of manual hardware configuration.

Analysis :

Compute Virtualization

Traditional Datacenter:

- Physical servers managed individually.
- Slow deployment.

SDDC:

- Virtual machines created within minutes.
- Automatic resource allocation.
- Easy workload migration.

Benefit:

Higher flexibility.

Storage Virtualization :

Traditional:

- Storage devices managed separately.
- Difficult expansion.

SDDC:

- Multiple storage devices combined into one virtual storage pool.
- Easy scalability.
- Better utilization.

Benefit:

Faster storage management.

Network Virtualization :

Traditional:

- Physical switches require manual configuration.
- Network changes are time-consuming.

SDDC:

- Virtual networks created instantly.
- Software-based routing and firewall management.
- Faster deployment.

Benefit:

Improved network agility.

Overall Analysis :

Compared to traditional datacenters, SDDC offers:

- Faster provisioning
- Automation
- Better scalability
- Reduced operational cost
- Simplified management
- Improved disaster recovery

Recommendation :

Organizations with dynamic workloads should adopt SDDC because it supports rapid deployment, centralized management, and efficient resource utilization.

Conclusion :

Software Defined Datacenters improve agility by virtualizing compute, storage, and networking resources, enabling automated management, rapid deployment, and better scalability compared to traditional datacenters.

Q5.A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

ANSWER :

Introduction :

Multi-cloud federation allows users to access services across different cloud providers using a common identity. Identity mismatches occur when providers use incompatible authentication systems, leading to login failures and security issues.

Analysis :

Problem

Different cloud providers:

- Use different user databases.
- Support different authentication protocols.
- Have inconsistent user roles and permissions.

Result:

- Users cannot access services across clouds.
- Authentication failures occur.
- Security risks increase.

Secure Identity Design :

Implement Identity Federation using:

- Single Sign-On (SSO)
- Identity Provider (IdP)
- Service Provider (SP)
- Standard protocols such as SAML 2.0 or OAuth/OpenID Connect.

Benefits:

- One login for multiple clouds.
- Consistent identity management.
- Simplified user access.

Privacy Protection :

- Encrypt user credentials.
- Use Multi-Factor Authentication (MFA).
- Apply Role-Based Access Control (RBAC).
- Share only required user information.
- Maintain audit logs for monitoring.

Recommendation :

The organization should deploy a centralized Identity Provider integrated with all cloud providers, enforce SSO and MFA, and use standard federation protocols to ensure secure and seamless access.

Conclusion :

Identity federation resolves authentication mismatches by enabling a common identity across multiple cloud providers. A secure design using SSO, MFA, RBAC, and standard federation protocols improves security, privacy, and user experience.